Greenclassify : Deep Learning-Based Approach for Vegetable Image Classification

ABSTRACT

Greenclassify is a Deep Learning-based system designed to automatically classify vegetable images using Convolutional Neural Networks (CNNs). The system analyzes input images and categorizes them into predefined vegetable classes with high accuracy. This solution helps automate sorting processes in agricultural industries, improves quality control in supply chains, and enhances smart shelf management in retail stores. The model is integrated with a Flask web application to provide an interactive user interface for real-time predictions.

INTRODUCTION

Vegetable classification is traditionally performed manually in agricultural processing units and retail stores. This process is time-consuming, labor-intensive, and prone to human errors. With the advancement of Artificial Intelligence and Deep Learning, image classification systems can now automatically recognize objects with high precision.

Greenclassify leverages Convolutional Neural Networks (CNN) to identify and classify vegetables from images. The system integrates a trained model with a Flask-based web application, enabling users to upload images and receive instant predictions.

PROBLEM STATEMENT

Manual vegetable sorting and inspection:

- Requires significant manpower
- Is prone to misclassification
- Reduces operational efficiency
- Causes delays in supply chain management

An automated deep learning-based classification system is required to improve accuracy, efficiency, and scalability.

OBJECTIVES

By the end of this project, we aim to:

- Understand fundamental concepts of Convolutional Neural Networks (CNNs)
- Gain knowledge of image data handling and feature extraction
- Perform data preprocessing techniques such as resizing and normalization
- Build and train a CNN model for vegetable classification
- Evaluate model performance using accuracy and other metrics

- Develop a Flask-based web application
- Deploy the trained model for real-world usage

EXISTING SYSTEM

The existing system relies on:

- Manual inspection
- Human-based sorting
- Basic image tagging without AI

Limitations:

- Time-consuming
- Error-prone
- Not scalable
- Inconsistent quality control

PROPOSED SYSTEM

The proposed system is an automated vegetable classification system using Deep Learning.

Features:

- Image-based vegetable detection
- CNN-based feature extraction
- Automatic classification
- Real-time prediction using Flask web application
- User-friendly interface

SYSTEM ARCHITECTURE

User → Upload Image → Flask Application
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Image Preprocessing (Resize, Normalize)
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CNN Model
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Feature Extraction
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Classification (Softmax Output)

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Display Prediction on UI

PROJECT FLOW

1. The user interacts with the User Interface (UI).
2. The user selects and uploads an image.
3. The image is sent to the Flask backend.
4. The model preprocesses the image.
5. CNN analyzes the image and extracts features.
6. Prediction is generated.
7. The predicted vegetable name is displayed on the UI.

METHODOLOGY

1. Data Collection

- Dataset collected from online sources (e.g., Kaggle).
- Images categorized into multiple vegetable classes.
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2. Data Preprocessing

- Resize images (224x224 pixels).
- Normalize pixel values.
- Split dataset into training and testing sets.
- Apply data augmentation using ImageDataGenerator.

3. Model Building

- Import necessary libraries.
- Define input shape.
- Add CNN layers.
- Compile model using:
    - Optimizer: Adam
    - Loss Function: Categorical Crossentropy
    - Metrics: Accuracy

4. Model Training

- Train model on training dataset.
- Monitor validation accuracy.
- Use augmentation to prevent overfitting.

5. Model Evaluation

- Evaluate model on test dataset.
- Calculate:
    - Accuracy
    - Confusion Matrix
    - Precision
    - Recall
    - F1-score

6. Model Deployment

- Save trained model (.h5 file).
- Integrate model into Flask application.
- Deploy locally on browser.

CNN ALGORITHM

The CNN architecture consists of:

- Convolution Layer – Extracts image features
- ReLU Activation – Introduces non-linearity
- Pooling Layer – Reduces spatial dimensions
- Flatten Layer – Converts 2D features into 1D
- Dense Layer – Fully connected layer
- Softmax Output – Produces classification probabilities

MODULES

1. Dataset Module – Handles image dataset.
2. Preprocessing Module – Cleans and prepares images.
3. Training Module – Builds and trains CNN model.
4. Prediction Module – Performs real-time classification.
5. Output Display Module – Shows results on UI.

TOOLS USED

- Python

- TensorFlow
- Keras
- OpenCV
- Flask
- HTML
- CSS

APPLICATIONS

- Vegetable Processing Units
- Retail Stores
- Smart Agriculture Systems
- Automated Sorting Systems
- Supply Chain Quality Control

ADVANTAGES

- High Accuracy
- Reduces Human Effort
- Saves Time
- Scalable Solution
- Real-Time Prediction
- Improves Inventory Management

RESULT

The system successfully classifies vegetable images with high accuracy. The trained CNN model integrated with Flask provides real-time predictions through a web-based interface.

CONCLUSION

Greenclassify demonstrates the effectiveness of Deep Learning in automating vegetable classification. By integrating CNN with a Flask web application, the system provides an efficient and scalable solution for agricultural and retail industries.

FUTURE SCOPE

- Freshness Detection
- Disease Detection in Vegetables
- Mobile Application Integration
- Real-Time Camera-Based Detection
- Cloud Deployment
- Multi-object detection using advanced models